

LIÈGE université
Sciences Appliquées

Transformers, power flow analysis part 2

ELEC0447 - Analysis of electric power and energy systems

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Overview I

1. The Transformer
2. Per unit representation
3. Transformers’ parameters, in practice
4. Tap-changing transformers and auto-transformers
5. Phase shifting
6. Transformers in the power flow analysis

What will we learn today?

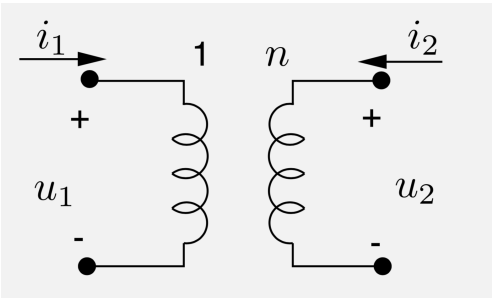
- The power transformer: physical models, nameplate test data ($v_k\%$, $v_{kr}\%$), and per-unit representation
- Delta-wye connections, phase shifts, and standard vector groups (IEC clock convention)
- Tap changers (OLTC) and phase-shifting transformers (PST) for voltage and power flow control
- How to include transformers in power flow analysis:
  - Two-port admittance matrix and system $\mathbf{Y}_{\text{bus}}$ stamping
  - Equivalent π-model for in-phase regulation ($t \in \mathbb{R}$)
  - Loss of $\mathbf{Y}_{\text{bus}}$ symmetry for phase shifters ($t \in \mathbb{C}$)
- Hands-on 4-bus case study with PST in PandaPower and Newton-Raphson

You will be able to do exercises 6.2, 6.3, 6.4, 6.18, 6.19 from Ned Mohan's book.

The Transformer

Basic model of a transformer

A (single phase) transformer is made of two magnetically coupled **coils** or **windings**. An ideal transformer is a two-port represented as



with

$$u_2 = nu_1$$

$$i_2 = -\frac{1}{n} i_1$$

Sign convention note: Currents i_1 and i_2 both enter the positive terminals (standard two-port convention), explaining the minus sign. With load convention (i_2 leaving), $i_2 = \frac{1}{n} i_1$.

Usages

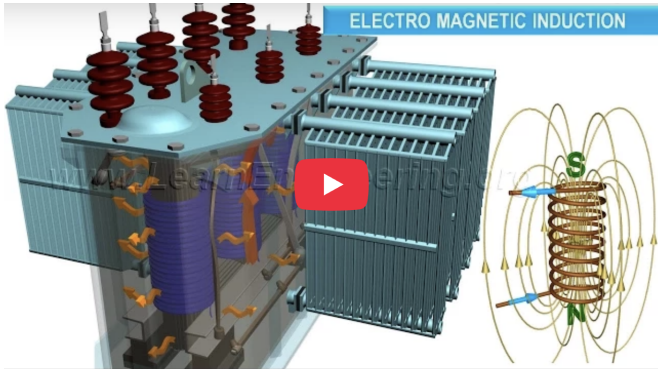
In power systems, transformers are mainly used to transmit power over long distances by changing the voltage level, thus decreasing the current for a given power level. The voltage level of a synchronous generator is around 20kV.

Voltage is changed around five times between generation and load.

It is also used to

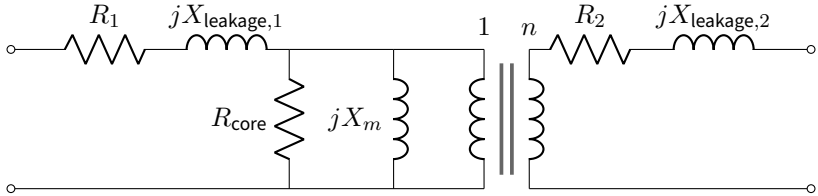
- **measure** currents and voltages,
- electrically **isolate** parts of a circuit (not the auto-transformer we will see),
- and **match** impedances.

How does a transformer work?



Link to the video

Non-ideal model I



Non-ideal model II

The ideal model is complemented by elements

- X_m that models the magnetizing inductance
- $X_{leakage,i}$ that models the flux not captured by the core on side i
- R_{core} that models eddy current and hysteresis losses, i.e., losses in the iron core
- R_1 and R_2 that model (coil) copper losses

Parameters are either given in the datasheet or obtained by open-circuit and short-circuit tests.

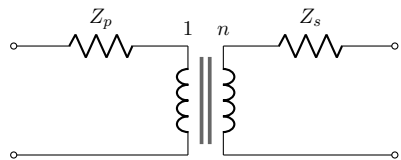
The core is usually **laminated** to decrease losses.

Simplification: Neglecting the Excitation Branch

Under rated operating conditions, the excitation current $\bar{I}_0$ (magnetizing + core losses) is very small:

$$\bar{I}_0 \approx 0.5\% \text{ to } 2\% \text{ of rated current}$$

- For power flow and short-circuit studies, the shunt branch ($R_{\text{core}} \parallel jX_m$) is **neglected**.
- The model reduces to two series winding impedances $Z_p = R_1 + jX_{\text{leakage},1}$ and $Z_s = R_2 + jX_{\text{leakage},2}$ separated by an ideal transformer $(1 : n)$.



Moving Impedances: Referring to the Primary Side

Refering to the circuit on the previous slide, the voltage seen from primary is:

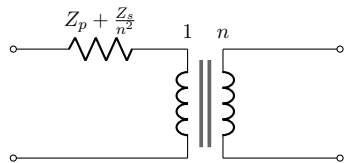
$$\bar{V}_p = Z_p \bar{I}_p + \frac{1}{n}\left(Z_s \bar{I}_s + \bar{V}_s\right) = \bar{I}_p \left(Z_p + \frac{1}{n^2} Z_s\right) + \frac{\bar{V}_s}{n}$$

since with turns ratio $1 : n$, $\bar{I}_p = n\bar{I}_s \iff \bar{I}_s = \bar{I}_p / n$.

- The total impedance seen from primary is then

$$Z_{\text{total}}^{(p)} = Z_p + \frac{Z_s}{n^2}$$

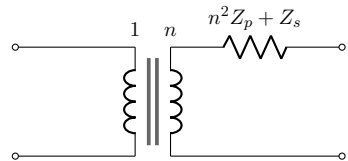
- We can remove the impedance Z_s from the secondary and place an impedance $\frac{Z_s}{n^2}$ on the primary side.



Moving Impedances: Referring to the Secondary Side

Exercise

Show that we can remove the impedance Z_p from the primary and place an impedance $n^2 Z_p$ on the secondary side.



Physical Intuition: Power Invariance

Conservation of Apparent Power Loss

$$S_{\text{loss}} = I_s^2 Z_s = \left(\frac{I_p}{n}\right)^2 Z_s = I_p^2 \left(\frac{Z_s}{n^2}\right) = I_p^2 Z_s'$$

Intuition from voltage and current levels:

- In a step-up transformer ($n > 1$), secondary has higher voltage and **lower current**.
- For the same power loss or relative drop, a smaller current requires a **larger ohmic impedance** ($Z \propto V^2/S$).
- **Rule of thumb for referral:**
  - Low-voltage $\rightarrow$ High-voltage: **multiply by** n^2 (impedance increases)
  - High-voltage $\rightarrow$ Low-voltage: **divide by** n^2 (impedance decreases)

Per unit representation

Per-Unit representation

Per-unit base selection for transformers

In power system analysis, base voltages across the transformer are chosen to match the nominal turns ratio:

$$V_{s,\text{base}} = nV_{p,\text{base}} \qquad \text{and} \qquad S_{p,\text{base}} = S_{s,\text{base}} = S_{\text{base}}$$

Base Currents:

$$I_{p,\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{p,\text{base}}} = nI_{s,\text{base}}$$

Base currents automatically satisfy the turns ratio!

Base Impedances:

$$Z_{s,\text{base}} = \frac{V_{s,\text{base}}^2}{S_{\text{base}}} = \frac{(nV_{p,\text{base}})^2}{S_{\text{base}}} = \mathbf{n^2 Z_{p,\text{base}}}$$

Ratio of impedance bases is **exactly** n^2!

Per-Unit Representation: Impedance Invariance

Let us convert the total referred impedance into per unit on each side:

1. From primary side (divided by $Z_{p,\text{base}}$):

$$Z_{\text{tr, pu}}^{(p)} = \frac{Z_{\text{total}}^{(p)}}{Z_{p,\text{base}}} = \frac{Z_p + Z_s/n^2}{Z_{p,\text{base}}} = \frac{Z_p}{Z_{p,\text{base}}} + \frac{Z_s}{n^2 Z_{p,\text{base}}} = \frac{Z_p}{Z_{p,\text{base}}} + \frac{Z_s}{Z_{s,\text{base}}}$$

2. From secondary side (divided by $Z_{s,\text{base}}$):

$$Z_{\text{tr, pu}}^{(s)} = \frac{Z_{\text{total}}^{(s)}}{Z_{s,\text{base}}} = \frac{n^2 Z_p + Z_s}{Z_{s,\text{base}}} = \frac{n^2 Z_p}{n^2 Z_{p,\text{base}}} + \frac{Z_s}{Z_{s,\text{base}}} = \frac{Z_p}{Z_{p,\text{base}}} + \frac{Z_s}{Z_{s,\text{base}}}$$

The per-unit impedance is strictly identical seen from either side:

$$\mathbf{Z}_{\text{tr, pu}} = \frac{\mathbf{Z}_{\text{total}}^{(\mathbf{p})}}{\mathbf{Z}_{\mathbf{p},\text{base}}} = \frac{\mathbf{Z}_{\text{total}}^{(\mathbf{s})}}{\mathbf{Z}_{\mathbf{s},\text{base}}} = \frac{\mathbf{Z}_{\mathbf{p}}}{\mathbf{Z}_{\mathbf{p},\text{base}}} + \frac{\mathbf{Z}_{\mathbf{s}}}{\mathbf{Z}_{\mathbf{s},\text{base}}}$$

Per-Unit Representation: Key Consequences

1. **Disappearance of Ideal Transformer ($1:1$):**

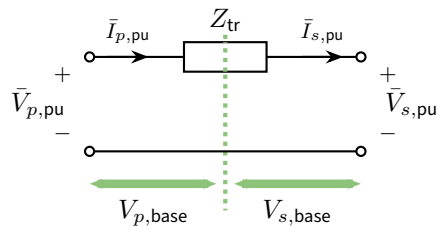
   $$\bar{V}_{s,\text{pu}} = \bar{V}_{p,\text{pu}}, \qquad \bar{I}_{p,\text{pu}} = \bar{I}_{s,\text{pu}}$$

   The turns ratio becomes $1:1$ and **completely vanishes**.

2. **Single Series Branch in Y_{bus}:** The transformer is modeled simply as a series branch $Z_{\text{tr, pu}}$ between buses.

3. **No Referral Ambiguity:** $Z_{\text{tr, pu}}$ is unique (no need to specify "HV" or "LV").

4. **Direct Link to Factory Data ($v_k\%$):**
   $v_k\% = 100 \times Z_{\text{tr, pu}}$, no need to separate Z_p, Z_s.



General Per-Unit Base Conversion Formula

When converting an impedance from $(V_{\text{base}}^{\text{old}}, S_{\text{base}}^{\text{old}})$ to a new base $(V_{\text{base}}^{\text{new}}, S_{\text{base}}^{\text{new}})$:

$$Z_{\text{pu}}^{\text{new}} = Z_{\text{pu}}^{\text{old}} \times \left(\frac{S_{\text{base}}^{\text{new}}}{S_{\text{base}}^{\text{old}}} \right) \times \left(\frac{V_{\text{base}}^{\text{old}}}{V_{\text{base}}^{\text{new}}} \right)^2$$

- When voltage bases across transformer zones match the nominal turns ratio:

$$\frac{V_{\text{base},1}}{V_{\text{base},2}} = \frac{V_{n1}}{V_{n2}} \implies \frac{V_{\text{base}}^{\text{old}}}{V_{\text{base}}^{\text{new}}} = 1$$

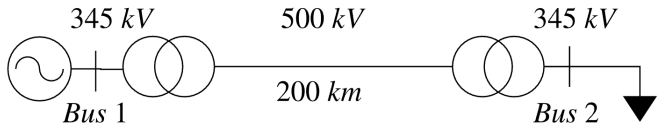
- Then Z_{pu} scales **solely with the MVA base ratio**:

$$Z_{\text{pu}}^{\text{new}} = Z_{\text{pu}}^{\text{old}} \times \left(\frac{S_{\text{base}}^{\text{new}}}{S_{\text{base}}^{\text{old}}} \right)$$

  This property is what makes per-unit analysis so powerful across multiple voltage tiers.

Illustration of Per-Unit normalization I

This is Example 6.1 from the reference book. Consider the one-line diagram



with

- a 200 km line with $R = 0.029\Omega/km$, $X = 0.326\Omega/km$, neglected shunt impedances
- two transformers with a leakage reactance of $0.2pu$ in the (500 kV, 1000 MVA) base, and losses neglected.

Illustration of Per-Unit normalization II

What is the equivalent model in a (345 kV, 100 MVA) base?

In the (500 kV, 1000 MVA) base:

- $Z_{line,pu} = 200 \times (0.029 + j0.326)/(500^2/1000) = 0.0232 + j0.2608pu$
- hence, the total impedance between buses 1 and 2 is

$$Z_{12} = 0.0232 + j0.2608 + 2 * j0.2pu = 0.0232 + j0.6608pu$$

In the (345 kV, 100 MVA) base

- the pu value of the impedance is the same in the (500 kV, 1000 MVA) and (345 kV, 1000 MVA) bases!

Illustration of Per-Unit normalization III

- Why? since we can transfer the impedance from one side of each transformer to the other, cf. a previous remark.
- if we now change the MVA base to 100 MVA,

$$Z_{12} = (0.0232 + j0.6608) \times (100/1000)pu = 0.00232 + j0.06608pu$$

since the base impedance is proportional to the inverse of the MVA base.

Transformers’ parameters, in practice

Bridging Physics to Datasheets: Why Nameplate Data?

The **nameplate** (*plaque signalétique*) is the certified metal rating plate affixed to the transformer tank, giving guaranteed ratings and factory test values under international standards (IEC 60076 / IEEE C57).

Total Referred Series Impedance is the Universal Link

$$Z_k \equiv Z_{\text{total}}^{(p)} = Z_p + \frac{Z_s}{n^2} \qquad \Longleftrightarrow \qquad z_k \equiv Z_{\text{tr, pu}} = \frac{v_k\%}{100}$$

(subscript k from German Kurzschluss / French court-circuit → short-circuit)

Why not the 6 physical parameters?

- **Impractical to separate:** internal parameters (R_1, R_2, $X_{\text{leakage},1}$, $X_{\text{leakage},2}$) are sealed in the tank.
- **Aggregated value suffices:** external grid flow and short circuits depend solely on total impedance Z_k.

Two Factory Acceptance Tests (IEC 60076):

- **Short-Circuit Test:** low voltage, secondary shorted → core flux negligible → **isolates series impedance** Z_k ($v_k\%$, P_{sc}).
- **Open-Circuit Test:** rated voltage, secondary open → series drop negligible → **isolates shunt branch** (P_{fe}, $i_0\%$).

Short-Circuit Test $\rightarrow$ Series Parameters (r_k, x_k, z_k)

Test Setup (IEC 60076):

- Secondary terminals **short-circuited** ($\bar{V}_2 = 0$).
- Voltage V_{sc} applied until rated current I_{n1} flows.
- Since $V_{\text{sc}} \approx 8\%$–$18\% V_n$, core flux is tiny $\rightarrow$ excitation current $\bar{I}_0 \approx 0$.
- Applied voltage drops **entirely across the series windings**:

$$Z_k = \frac{V_{\text{sc}}}{\sqrt{3} I_{n1}} = (R_1 + \frac{R_2}{n^2}) + j(X_{\text{leakage},1} + \frac{X_{\text{leakage},2}}{n^2}) \equiv Z_{\text{total}}^{(p)}$$

Direct Link to Nameplate Data

- **Impedance voltage $v_k\%$:**

$$z_{k,\text{tr}} = \frac{Z_k}{Z_n} = \frac{V_{\text{sc}}}{V_n} = \frac{\mathbf{v_k}\%}{\mathbf{100}}$$

- **Copper losses P_{sc} or $v_{kr}\%$:**

$$r_{k,\text{tr}} = \frac{R_k}{Z_n} = \frac{P_{\text{sc}}}{S_n} = \frac{\mathbf{v_{kr}}\%}{\mathbf{100}}$$

- **Total leakage reactance x_k:**

$$x_{k,\text{tr}} = \sqrt{\mathbf{z}_{\mathbf{k,tr}}^{\mathbf{2}} - \mathbf{r}_{\mathbf{k,tr}}^{\mathbf{2}}}$$

Open-Circuit Test $\rightarrow$ Shunt Parameters (R_{core}, X_m)

Test Setup (IEC 60076):

- Secondary terminals **open-circuited** ($I_2 = 0$).
- Rated voltage V_n applied to primary.
- Current is only no-load excitation current $\bar{I}_0$ ($< 0.5\%$ of I_n).
- Series drops ($I_0 R_1, I_0 X_{\text{leakage},1}$) are negligible $\rightarrow$ rated voltage falls **entirely across the shunt branch** ($R_{\text{core}} \parallel jX_m$).

Link to Nameplate Data

- **Iron / core losses P_{fe}:**

$$P_{\text{fe}} = \frac{V_n^2}{R_{\text{core}}} \implies g_{\text{fe,tr}} = \frac{\mathbf{P_{\text{fe}}}}{\mathbf{S_n}} \text{ [pu]}$$

- **No-load current $i_0\%$:**

$$I_0 \approx \frac{V_n/\sqrt{3}}{X_m} \implies b_{m,\text{tr}} \approx \frac{\mathbf{i_0\%}}{\mathbf{100}} \text{ [pu]}$$

Neglected in Power Flow

Since $i_0\% \leq 0.5\% \implies x_m \geq 200$ pu, shunt branch is usually omitted in transmission grid studies.

Summary: Model Mapping & PandaPower Implementation

Physical Element	Factory Test / Source	Nameplate Parameter	PandaPower
$R_1 + R_2/n^2$	Short-circuit test	P_{sc} [kW] or $v_{kr}\%$	`vkr_percent`
$X_{\text{leakage},1} + X_{\text{leakage},2}/n^2$	Short-circuit test	$v_k\%$	`vk_percent`
R_{core} (iron losses)	Open-circuit test	P_{fe} [kW] (or $p_0\%$)	`pfe_kw`
X_m (magnetizing)	Open-circuit test	$i_0\%$	`i0_percent`
$1 : n$ (turns ratio)	Winding ratio	V_{n1}, V_{n2} [kV]	`vn_hv_kv, vn_lv_kv`
Rating base	Rated capacity	S_n [MVA]	`sn_mva`

Base Conversion to Network Base ($S_{\text{base}} = \texttt{net.sn_mva}$)

$$\texttt{z_k} = \frac{\texttt{vk_percent}}{100} \times \frac{S_{\text{base}}}{S_n}, \quad \texttt{r_k} = \frac{\texttt{vkr_percent}}{100} \times \frac{S_{\text{base}}}{S_n}, \quad \texttt{x_k} = \sqrt{\texttt{z_k}^2 - \texttt{r_k}^2}$$

Field Example: 3-Phase Distribution Transformer (Dakar)



Picture from B. Cornélusse. Senelec, Dakkar.

What can you deduce from the exterior?

Field Example: 3-Phase Distribution Transformer (Dakar)



Picture from B. Cornélusse. Senelec, Dakkar.

What can you deduce from the exterior?

- **High-Voltage (MV) side (right):** 3 tall porcelain bushings with creepage skirts → needs high insulation distance; low current ($I \approx 8$ A).
- **Low-Voltage (LV) side (left):** 4 heavy flat copper busbars with bolt holes → high current ($I > 500$ A); needs large contact area.
- **Why 4 LV terminals?** 3 phases + neutral (a, b, c, n) → **secondary is Star/Wye (Y with neutral)** for 400 V 3-phase / 230 V 1-phase!
- **Center mechanism on top cover:** Off-circuit manual tap changer dial (5 positions).
- **Cooling:** Corrugated cooling fins → oil-immersed, natural convection (**ONAN**).

Field Example: Reading the Nameplate



Confirmation from the Nameplate:

- **Rated Power (S_n):** $\mathbf{400}$ kVA (0.4 MVA), 50 Hz, 3-phase.
- **Rated Voltages:** $V_{n1} = \mathbf{30\,000}$ V (HT) $\rightarrow V_{n2} = \mathbf{400/230.94}$ V (BT).
- **Vector Group:** **Dyn11** Delta on MV (3 bushings); Wye/Star on LV with neutral brought out (a, b, c, n); $30°$ phase advance.
- **Rated Currents:** $I_{n1} = \mathbf{7.70}$ A (HT) vs. $I_{n2} = \mathbf{577.4}$ A (BT)! *(Explains the heavy flat copper busbars on the LV side!)*
- **Tap Changer (HT):** 5 positions ($\pm 2 \times 2.5\%$): 31.5 kV $\leftrightarrow$ 30.75 kV $\leftrightarrow$ $\mathbf{30}$ kV $\leftrightarrow$ 29.25 kV $\leftrightarrow$ 28.5 kV
- **Cooling & Masses:** ONAN; oil 285 kg; total mass 1285 kg.

Field Example: Deriving the Grid Model Parameters

Factory Test Data → Per-Unit

Nameplate measurements:

$S_n = 400$ kVA, $V_{n1} = 30$ kV, $V_{n2} = 400$ V

$v_k\% = \mathbf{4.54}\%, \quad P_k = \mathbf{6437}$ W, $\quad P_o = \mathbf{738}$ W

Per-Unit Model ($S_{\text{base}} = S_n$):

$z_{k,\text{tr}} = \frac{v_k\%}{100} = \mathbf{0.0454}$ pu

$r_{k,\text{tr}} = \frac{P_k}{S_n} = \frac{6437}{400\,000} = \mathbf{0.0161}$ pu $(v_{kr}\% = 1.61\%)$

$x_{k,\text{tr}} = \sqrt{z_k^2 - r_k^2} = \mathbf{0.0425}$ pu (4.25%)

$g_{\text{fe}} = \frac{P_o}{S_n} = \frac{738}{400\,000} = \mathbf{0.00185}$ pu (0.18%)

Physical Ohms & PandaPower

Referred to HV (30 kV):

Base: $Z_{n,\text{HV}} = V_{n1}^2/S_n = 30^2/0.4 = \mathbf{2250\,\Omega}$

$R_k = r_k Z_{n,\text{HV}} = 0.0161 \times 2250 = \mathbf{36.2\,\Omega}$

$X_k = x_k Z_{n,\text{HV}} = 0.0425 \times 2250 = \mathbf{95.6\,\Omega}$

PandaPower Function Call:

```
pp.create_transformer_from_parameters(net, hv, lv,
  sn_mva=0.4, vn_hv_kv=30.0, vn_lv_kv=0.4,
  vk_percent=4.54, vkr_percent=1.61, pfe_kw=0.738,
  vector_group="Dyn11", shift_degree=30.0)
```

Vector group: defines zero-sequence paths (faults);

`shift_degree=30.0` applies the 30° shift in `runpp`.

Efficiency

The efficiency expressed in % is

$$100 \times \frac{P_{output}}{P_{input}} = 100 \times \left(1 - \frac{P_{losses}}{P_{input}}\right)$$

Remarks:

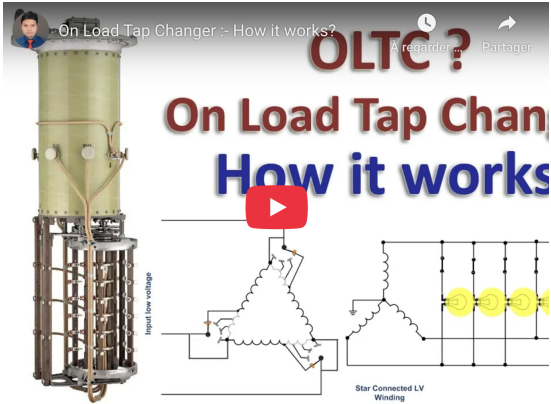
- maximal when loaded such that copper losses = iron losses (cancel derivative of efficiency w.r.t current)
- Around **99.5 %** in large power transformers at full load.

Tap-changing transformers and auto-transformers

Tap changers

- Some transformers are equipped with a system allowing the $1 : n$ ratio to be changed
- The ability to change the tap under load is called load tap changer (LTC) or on-load tap changer (OLTC)
- This is mainly used for voltage control
- It is usually implemented using auto-transformers
- We will see later on how to include this in the power flow analysis

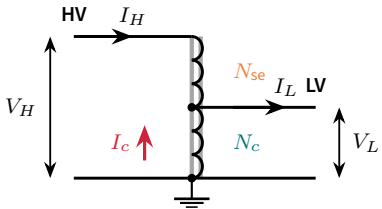
Tap changing transformers



Link to the video

Auto-transformers: Working Principle

Single continuous winding per phase on a shared magnetic core (no galvanic isolation).



Relations: $\dfrac{V_H}{V_L} = 1 + \dfrac{N_{se}}{N_c}, \qquad I_c = I_L - I_H$

Dual Power Transfer: ($S_{\text{rated}} = V_H I_H = V_L I_L$)

- **Inductive** (magnetic core flux):
  $$S_{\text{ind}} = (V_H - V_L) I_H = S_{\text{rated}} \left(1 - \frac{V_L}{V_H}\right)$$
- **Conductive** (direct metallic path):
  $$S_{\text{cond}} = V_L I_H = S_{\text{rated}} \left(\frac{V_L}{V_H}\right)$$

Key Result: Only S_{ind} crosses the core $\Longrightarrow$ the physical core is sized for S_{ind}, not the full throughput S_{rated}!

Auto-transformers sizing: why $V_H/V_L \approx 1$ matters

The **core sizing factor** (co-ratio) defines the fraction of power crossing the core:

$$k_{\text{size}} = \frac{S_{\text{ind}}}{S_{\text{rated}}} = 1 - \frac{V_L}{V_H} = \frac{V_H - V_L}{V_H} \qquad (k_{\text{size}} = 1 \text{ for 2-winding transformer})$$

When $V_H/V_L \approx 1 \implies$ High Savings ($k_{\text{size}} \ll 1$):

- **Grid tie ($380/220$ kV, $V_H/V_L \approx 1.73$):**
  $k_{\text{size}} = 1 - 220/380 = \mathbf{0.42}$
  $\rightarrow$ **1000 MVA** throughput needs only a **420 MVA** core! (58% saved in core size, weight, losses).
- **OLTC Regulation ($\pm 10\%$ boost, $V_H/V_L = 1.1$):**
  $k_{\text{size}} = 1 - 1/1.10 \approx \mathbf{0.09} \implies$ core handles only **9%**!

When $V_H/V_L \gg 1 \implies$ Negligible Savings:

- **Generator Step-Up ($20/400$ kV):**
  $k_{\text{size}} = 1 - 20/400 = \mathbf{0.95}$ (only 5% savings).
- Losing galvanic isolation is not worth a 5% saving $\rightarrow$ standard 2-winding used instead!

Rule of thumb: Auto-transformers are economically justified when $\mathbf{V_H/V_L \leq 2}$.

Auto-transformers: Trade-offs & Applications

Advantages:

- Physically smaller and lighter
- Less costly (significantly less copper & iron)
- Higher efficiency ($\eta > 99.7\%$)
- Lower leakage reactance
- Easy to implement tap changers (OLTC)
- Common solid earth ground

Disadvantages:

- No galvanic / electrical isolation
- Higher short-circuit current
- Risk of full V_H on secondary if common winding opens
- Economic benefit diminishes if ratio $V_H/V_L > 2$

Typical Applications: Interconnecting close transmission voltage levels (e.g. 380/220 kV, 220/150 kV) and serving as the regulating unit in on-load tap changers and phase shifters.

Phase shifting

Phase shifting

Transformers causing a phase shift

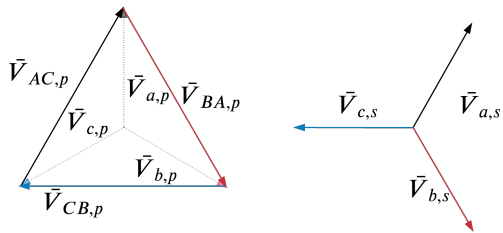
Some transformers naturally introduce a phase shift, as is explained below for *Dy* transformers.

- In radial systems, the phase shift has no consequence, as long as parallel transformers/paths introduce the same phase shift.
- In meshed networks, we can use this property to control power flows using dedicated *phase shifting transformers*.

Phase shift in delta-star transformers

The star part has n times the number of turns of the delta part (primary side).
Let's reason on phase a,

- Voltage $\bar{V}_{a,s}$ is on the same core as $\bar{V}_{AC,p} = \sqrt{3}\bar{V}_{a,p}\angle -30°$ where $\bar{V}_{a,p}$ is the (virtual) phase-neutral voltage on the primary side.
- Since $\bar{V}_{a,s} = n\bar{V}_{AC,p}$, $\bar{V}_{a,s} = n\sqrt{3}\bar{V}_{a,p}\angle -30°$



We **gain a $\sqrt{3}$ factor** in the amplification, and a **lagging phase shift of 30°**.

The same reasoning holds for phases b and c.

Standard Three-Phase Vector Groups (IEC 60076)

In industrial practice, phase displacement between windings is designated by a **clock number** (each hour represents a 30° lag):

Notation Rules:

- 1st letter: HV winding (**D** = Delta, **Y** = Wye)
- 2nd letter: LV winding (**d** = delta, **y** = wye, **n** = neutral out)
- Number $k \in \{0, \ldots, 11\}$: phase lag of LV line-to-neutral phasor relative to HV: $\theta_{\text{LV}} - \theta_{\text{HV}} = -k \times 30^\circ$

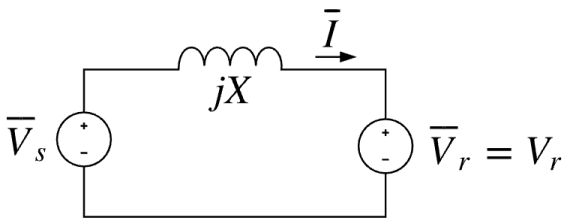
Common Examples:

- **Yy0 / Dd0**: 0° shift (12 o'clock)
- **Dy1**: Delta-Wye, LV lags by 30° (1 o'clock) — *the case studied on previous slide!*
- **Dy11** (or **Yd11**): LV leads by 30° / lags by 330° (11 o'clock)
- **Yd5**: LV lags by 150° (5 o'clock)

Paralleling Condition: Transformers can be operated in parallel only if they belong to the **same vector group** (identical phase displacement).

Power flow regulation by phase shifting I

We have seen that active power flows are dictated by the voltage magnitudes but also the sine of the angle difference between buses:



$$S_r = \bar{V}_r \bar{I}^* = V_r \left(\frac{V_s \angle -\theta - V_r}{-jX} \right)$$

$$= \frac{V_s V_r \sin\theta}{X} + j \frac{V_s V_r \cos\theta - V_r^2}{X}$$

θ is the angle between $\bar{V}_r$ and $\bar{V}_s$

Power flow regulation by phase shifting II

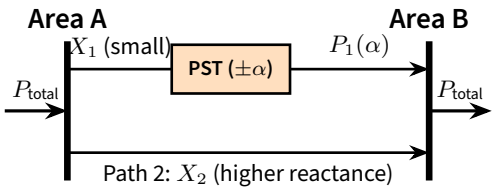
In per unit ($V_s \approx V_r \approx 1$), if the angle difference is small (DC approximation), we have

$$P_r = \frac{\theta_r - \theta_s}{X}.$$

If we have a device that can generate an adjustable phase shift, we can control the power flows. This is the purpose of **phase-shifting transformers**.

Physical Principle: Controlling Parallel Path & Loop Flows

Active power divides inversely proportional to line reactances: $P \sim \Delta\theta / X$. In meshed grids, this creates **transit flows** and line overloads.



Without PST ($\alpha = 0$) (demonstrate this relation):

$$P_1 \approx P_{\text{total}} \frac{X_2}{X_1 + X_2} \quad \text{(overload on Line 1!)}$$

With PST ($\alpha \neq 0$):

$$P_1 \approx \frac{\theta_A - \theta_B + \alpha}{X_1}$$

Tuning α redirects power into Path 2, relieving Line 1 without costly generation redispatch!

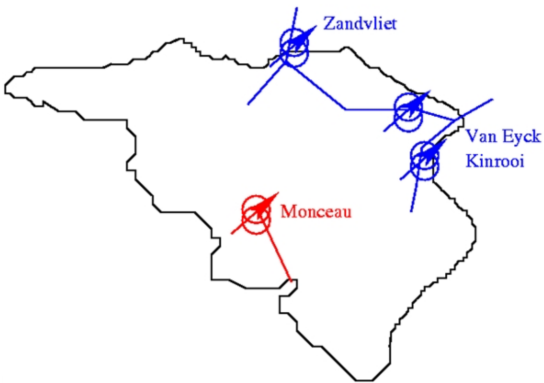
Why PSTs on Belgian borders? German wind power flowing to Southern Germany creates large transit loop flows across Belgium/Netherlands. PSTs act as electrical valves to protect interconnectors.

Python example: parallel path and loop flows

Let’s illustrate these notions and look at the difference between AC power flow and DC power flow:

▸ PandaPower Notebook

Phase shifting transformers on the borders of Belgium I



380/380 kV, in series with:

1. line Zandvliet (B) - Borssele (NL) and Zandvliet (B) - Geertruidenberg (NL)
2. line Meerhout (B) - Maasbracht (NL)
3. line Gramme (B) - Maasbracht (NL)
   - nominal power $3V_N I_{\max} = 1400$ MVA
   - phase shift adjustment: 35 positions, $+17/-17 \times 1.5°$ (at no load)

Phase shifting transformers on the borders of Belgium II

220/150 kV :

- in series with the Chooz (F) - Monceau (B) line nominal power: 400 MVA
- in-phase adjustment: 21 positions, $+10/-10 \times 1.5\%$
- quadrature adjustment: 21 positions, $+10/-10 \times 1.2°$

Phase-Shifting Transformers: Quadrature Injection Principle

Physical Principle: 90° Quadrature Injection

- **Objective**: Steer active power ($P \approx \Delta\theta / X$) by adding a series quadrature voltage:

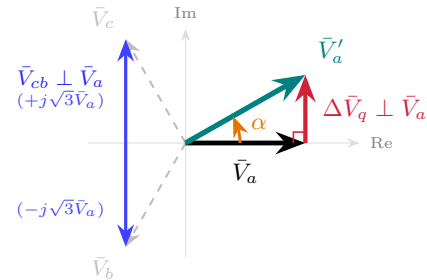
$$\bar{V}_a' = \bar{V}_a + \Delta\bar{V}_q \quad (\text{with } \Delta\bar{V}_q \perp \bar{V}_a)$$

- **Natural 90° Source**: In balanced 3-phase, the delta voltage across the other two phases is **orthogonal** to phase a:

$$\bar{V}_{bc} = \bar{V}_b - \bar{V}_c = -j\sqrt{3}\,\bar{V}_a \implies \bar{V}_{bc} \perp \bar{V}_a$$

- **Phase Shift Control**: Tapping a controllable fraction $\pm k$ gives:

$$\tan\alpha = \frac{|\Delta\bar{V}_q|}{|\bar{V}_a|} \approx \pm\sqrt{3}k \quad [\text{rad}]$$



Purely Electromagnetic: Standard AC windings and tap changer provide the shift without power electronics!

Further reading: Weedy §5.7; ENTSO-E PST Modelling Guide.

Transformers in the power flow analysis

Transformer without regulation

A transformer, in the per-unit representation, can thus be represented

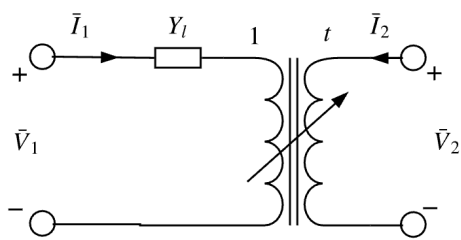
- as a two-port if the shunt admittance is considered
- as a simple series leakage impedance if the shunt admittance is neglected

It is a bit harder to model controllable taps for voltage and flow control.

Representing taps and phase shifts I

Let Y_l be the leakage admittance and t be the off-nominal turns ratio:

- If $t \in \mathbb{R}$ (typically $0.90 \leq t \leq 1.10$, step size $\approx 0.5\%$–1.25%), this corresponds to an **in-phase tap-changer (OLTC)** for voltage magnitude and reactive power control.
- If $t = me^{j\theta}$ with $\theta \neq 0$ ($|\theta| \leq 15°$–$25°$), this is a **phase-shifting transformer (PST)** for active power flow control.



Representing taps and phase shifts II

We have

$$\bar{I}_1 = \left(\bar{V}_1 - \frac{\bar{V}_2}{t}\right) Y_l$$

and since $\frac{\bar{V}_2}{t}\bar{I}_1^\star = -\bar{V}_2\bar{I}_2^\star$ by energy conservation

$$\bar{I}_2 = -\frac{\bar{I}_1}{t^\star} = -\bar{V}_1\frac{Y_l}{t^\star} + \bar{V}_2\frac{Y_l}{|t|^2}$$

Thus tap and phase shift can be represented by the two-port admittance matrix:

$$\begin{bmatrix}\bar{I}_1\\ \bar{I}_2\end{bmatrix} = \begin{bmatrix} Y_l & -\frac{Y_l}{t} \\ -\frac{Y_l}{t^\star} & \frac{Y_l}{|t|^2}\end{bmatrix}\begin{bmatrix}\bar{V}_1\\ \bar{V}_2\end{bmatrix}$$

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact I

For a transformer connected between **Bus** i (series impedance side) and **Bus** j (ideal tap side $1 : t$):

$$\begin{bmatrix} \bar{I}_i \\ \bar{I}_j \end{bmatrix} = \begin{bmatrix} Y_l & -\dfrac{Y_l}{t} \\ -\dfrac{Y_l}{t^\star} & \dfrac{Y_l}{|t|^2} \end{bmatrix} \begin{bmatrix} \bar{V}_i \\ \bar{V}_j \end{bmatrix}$$

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact II

Stamping Rule into the Global $\mathbf{Y}_{\text{bus}}$ Matrix:

$$
\begin{aligned}
Y_{ii} &\leftarrow Y_{ii} + Y_l \\
Y_{jj} &\leftarrow Y_{jj} + \frac{Y_l}{|t|^2} \\
Y_{ij} &\leftarrow Y_{ij} - \frac{Y_l}{t} \\
Y_{ji} &\leftarrow Y_{ji} - \frac{Y_l}{t^\star}
\end{aligned}
$$

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact III

Consequence for Complex Taps ($t = me^{j\theta}, \theta \neq 0$)

Because $t \neq t^{\star}$, the off-diagonal terms differ:

$$Y_{ij} \neq Y_{ji} \implies \mathbf{Y}_{\text{bus}} \neq \mathbf{Y}_{\text{bus}}^{T} \quad \text{(loss of symmetry!)}$$

Direct Impact on the Newton-Raphson Power Flow:

- Standard textbook formulations often assume $Y_{km} = Y_{mk}$ ($G_{km} = G_{mk}, B_{km} = B_{mk}$).

System $\mathbf{Y}_{\text{bus}}$ Stamping & Newton-Raphson Impact IV

- With PSTs, one **must** strictly use directed branch admittance values:

$$P_k = \sum_{m=1}^{n} V_k V_m \left(G_{km} \cos(\theta_k - \theta_m) + B_{km} \sin(\theta_k - \theta_m) \right)$$

  where $G_{km} + jB_{km} = Y_{km} \neq Y_{mk}$.

- Off-diagonal blocks in the Jacobian $\mathbf{J}$ are no longer transpose-symmetric.

Illustration: The 4-Bus System with PST I

Let's adapt our previous example by placing a PST between Bus 1 and an auxiliary Bus 4:

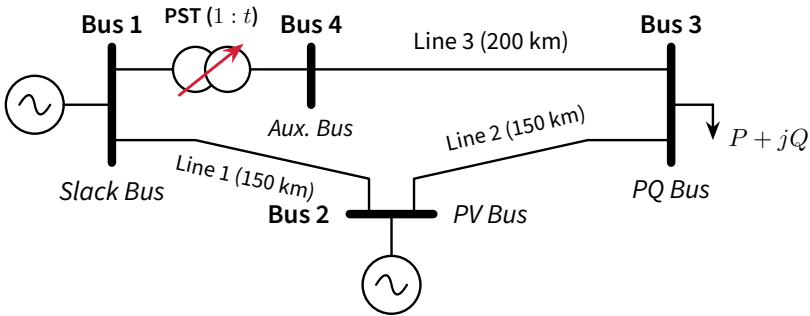


Illustration: The 4-Bus System with PST II

Python Notebooks & Implementation

Let's see how to adapt the admittance matrix, now of size 4×4

▸ PandaPower Notebook ▸ Newton-Raphson Script

Remark – Network Reduction (Kron / Norton Equivalent): Since no power is injected at auxiliary Bus 4 ($I_4 = 0$):

$$\mathbf{Y}_{\text{reduced}} = \mathbf{Y}_{aa} - \mathbf{Y}_{ab} Y_{44}^{-1} \mathbf{Y}_{ba}$$

This yields an exact 3×3 equivalent network for custom Newton-Raphson solvers.

Illustration: The 4-Bus System with PST III

Effect of PST Tap Shift (from $-8°$ to $+8°$):

Tap Angle	Line 1	Line 2	Line 3
$-8°$	61.2 %	43.1 %	**98.5 %**
$0°$	75.4 %	68.2 %	84.1 %
$+8°$	**92.8 %**	89.4 %	67.3 %

- As tap angle increases, active power is diverted from Line 3 towards Path 1–2.
- Line 3 is relieved without changing generator outputs!